

• LOGARITMI: CASI COMUNI

• $a = 2 \rightarrow \log_2$

• $a = 10 \rightarrow \log_{10}$

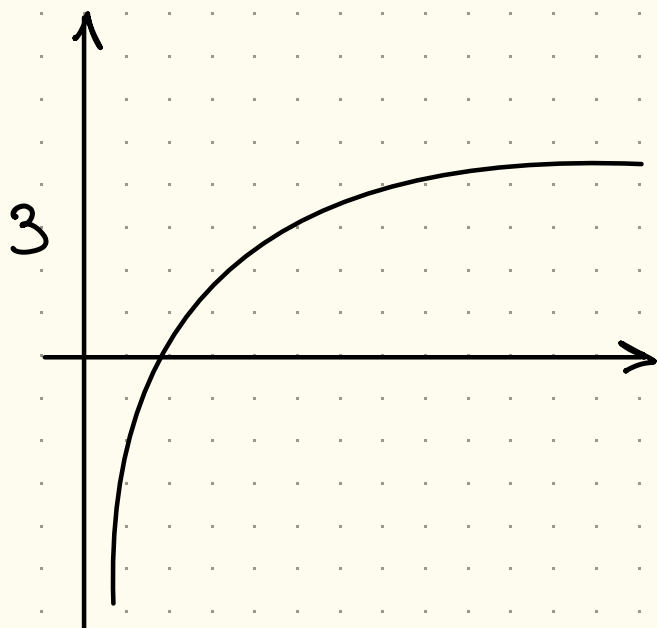
es) 10^x : $10^4 = 10.000$

$$\begin{array}{c} \Downarrow \\ \log_{10} 10.000 = 4 \end{array}$$

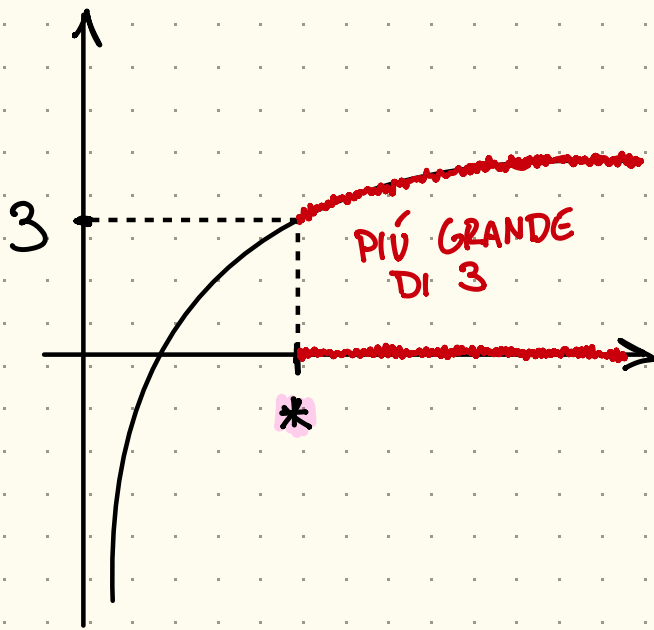
• $a = e \rightarrow \log_e = \log = \ln$

es) $\log x^2 > 3$

1. DISEGNO LA FUNZIONE $\log$



2. QUANDO IL log DI QUALCOSA È
MAGGIORE DI 3?



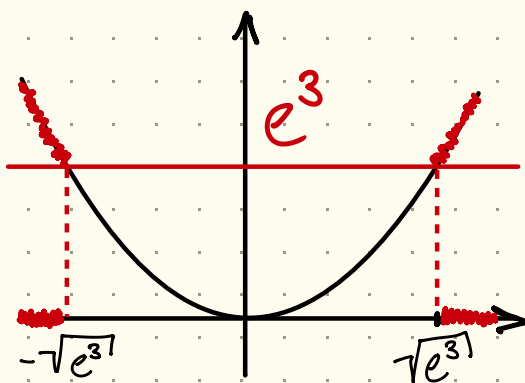
3. CHI È L'ASTERISCO?

$$\log x^2 > 3$$

$$\Downarrow$$
$$x^2 > *$$

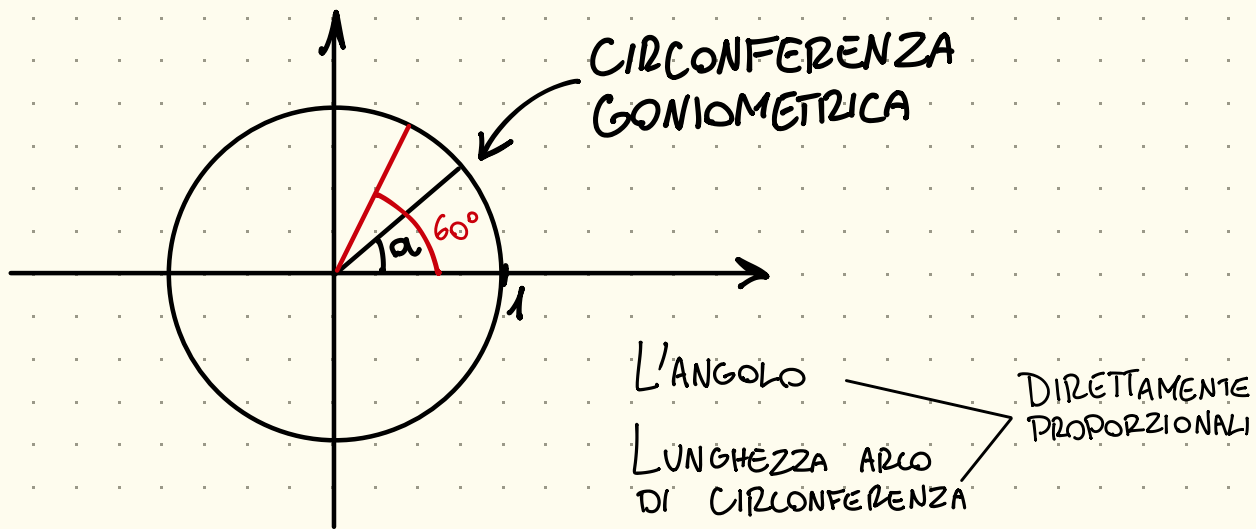
$$\Downarrow$$
$$* = e^3$$

$$\Downarrow$$
$$\log e^3 = 3$$



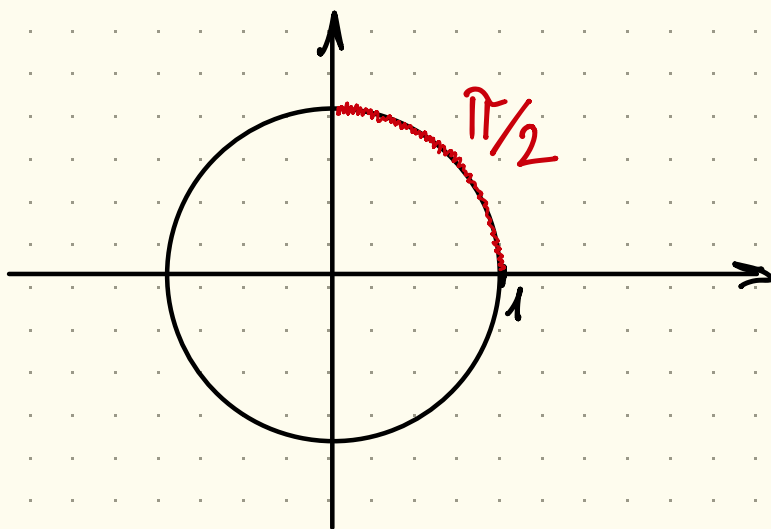
• TRIGONOMETRIA

- RADIANTI



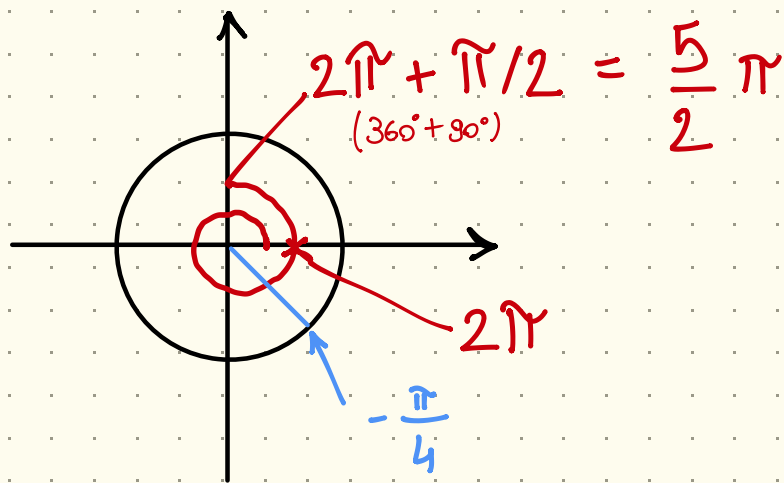
- GLI ANGOLI IN TRIGONOMETRIA SI MISURANO SEMPRE A PARTIRE DALL'ASSE X, IN SENSO ANTICLOCKWISE

- ANGOLO IN RADIANTI $\Rightarrow$ LUNGHEZZA



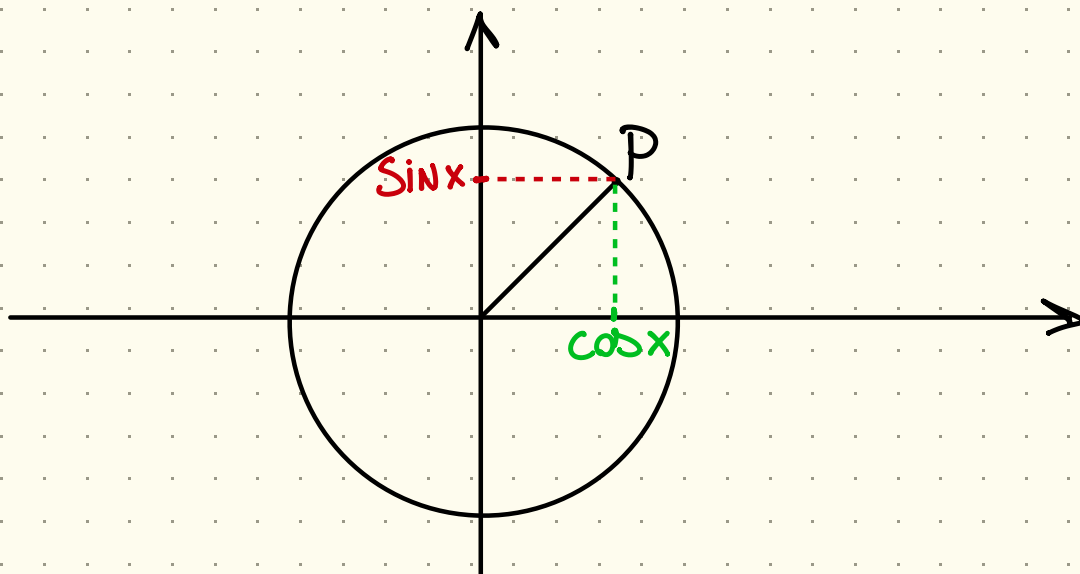
GRADI	RADIANTI
0	0
30	$\pi/6$
45	$\pi/4$
60	$\pi/3$
90	$\pi/2$
180	π
270	$3/2 \pi$
360	2π

- POSSO AVERE ANCHE ANGOLI NEGATIVI O
ANGOLI PIU' LUNGHI DI 2π



$$\frac{\alpha \text{ GRADI}}{360} = \frac{\alpha \text{ RADIANTI}}{2\pi}$$

• SENO e COSENO



$$P = (\cos(x), \sin(x))$$

• PROPRIETÀ'

$$\left. \begin{aligned} \bullet \sin(x + 2\pi) &= \sin x \\ \bullet \cos(x + 2\pi) &= \cos x \end{aligned} \right\} \forall x \in \mathbb{R}$$

$$\left. \begin{aligned} \bullet \sin(x + K \cdot 2\pi) &= \sin x \\ \bullet \cos(x + 2K\pi) &= \cos x \end{aligned} \right\} \begin{aligned} &\forall x \in \mathbb{R} \\ &\forall K \in \mathbb{Z} \Rightarrow K \text{ DEVE ESSERE UN NUMERO INTERO!!} \end{aligned}$$

→ SIN E COS SONO PERIODICHE

DI PERIODO 2π :

$$\left\{ \begin{aligned} - \text{Dom } \sin &= \mathbb{R} \\ - \text{Dom } \cos &= \mathbb{R} \end{aligned} \right.$$

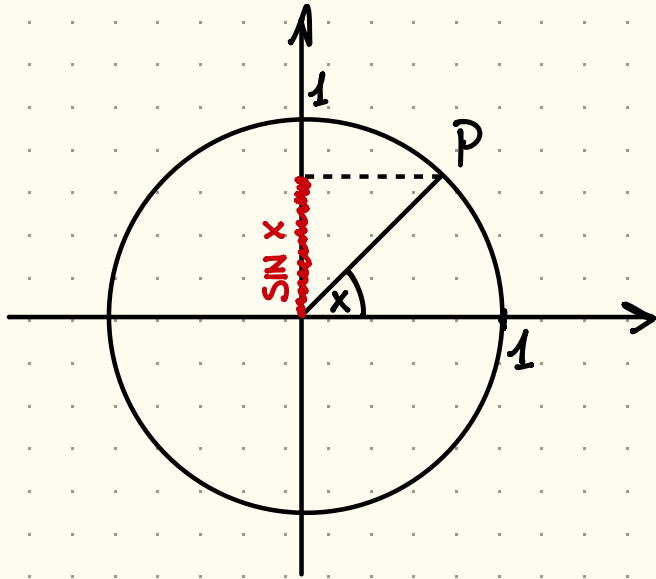
→ AD ESEMPIO LE ORE DEL GIORNO,
CON PERIODO 24, DOPO 24 ORE
SEI NELLO STESSO PUNTO DELL'OROLOGIO

• $\tan$ NON È DEFINITA IN:

$$\left\{ \begin{aligned} x &= \frac{\pi}{2} + 2K\pi \\ x &= \frac{3}{2}\pi + 2K\pi \end{aligned} \right\} \text{ CON } K \in \mathbb{Z}$$

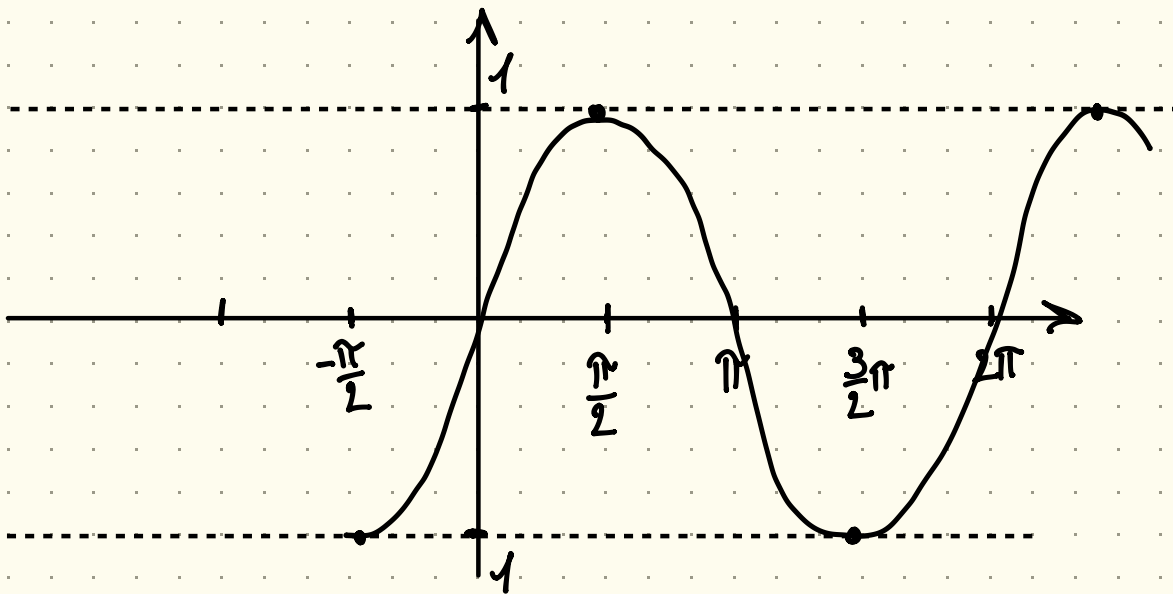
$$\bullet \tan x = \tan(x + K\pi) \quad \begin{aligned} &\forall x \in \text{Dom } \tan \\ &\forall K \in \mathbb{Z} \end{aligned}$$

• GRAFICO DEL SENO

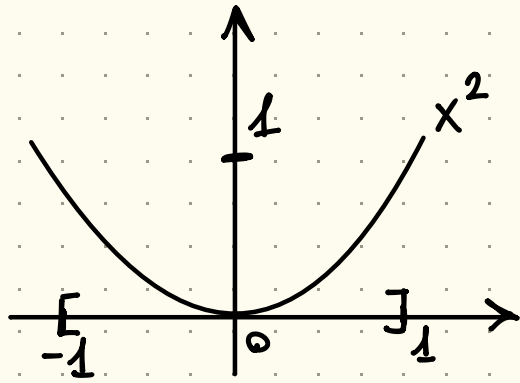


x	$\sin x$
0	0
$\frac{\pi}{4}$	
$\frac{\pi}{2}$	1
π	0
$\frac{3}{2}\pi$	-1
2π	0

• RAPPRESENTAZIONE VALORI



es) $f(x) = (\sin x)^2$



$$\text{Dom } f = \mathbb{R}$$

$$\text{Im } f = [0, 1]$$